

SITTING, STEPPING, AND ACCUMULATION PATTERNS ACROSS WORK-FROM-HOME AND WORKPLACE CONTEXTS IN ADULTS WITH TYPE 2 DIABETES

Lisa Becker^{1,2}, Genevieve N Healy³, Elisabeth A H Winkler³, Paul A Gardiner⁴, Ruth Grigg², Paddy C Dempsey^{1,2,5,6}, Michael J Wheeler^{1,2}, David W Dunstan^{1,2}

¹ Deakin University, Deakin Institute for Physical Activity and Nutrition, School of Exercise and Nutrition Sciences, Melbourne, Australia

² Baker Heart and Diabetes Institute, Melbourne, Australia

³ The University of Queensland, School of Human Movement and Nutrition Sciences, Brisbane, Australia

⁴ The University of Queensland, School of Public Health, Brisbane, Australia

⁵ Diabetes Research Centre, Leicester General Hospital, University of Leicester, Leicester, United Kingdom

⁶ Institute of Metabolic Science (IMS) Epidemiology, University of Cambridge, Cambridge Biomedical Campus, Cambridge, United Kingdom

Email: s222289893@deakin.edu.au

INTRODUCTION

Hybrid working arrangements have become more prolific, yet objective evidence on its association with movement behaviours remains limited, particularly among adults with type 2 diabetes, a population at high risk of adverse health effects from prolonged sedentary behaviour.

We investigated whether device-measured sitting, stepping, and sedentary accumulation patterns differed between work-from-home and workplace contexts during waking hours on workdays and during working hours, and explored hourly temporal patterns in office workers with type 2 diabetes.

METHODS

This secondary analysis included 71 non-intervention (control) participants from the *OPTIMISE Your Health* trial: adults aged 35-65 years with type 2 diabetes in predominantly desk-based occupations. Baseline, 3-month, and 6-month data collected between July 2019 and October 2024 were analysed. Sitting/lying time, prolonged sitting time (≥ 30 min bouts), mean sitting/lying bout duration, and stepping time were measured using activPAL4 monitors worn on the upper thigh. A concurrent diary (24-h, 10-day protocol) identified work-from-home and workplace contexts. Time-based outcomes were standardised to 16 waking hours and 8 working hours.

Mixed-effects models compared these contexts during waking hours on workdays ($n=1099$ days) and during working hours ($n=1034$ days), accounting for repeated measures, timepoint, confounders, and interactions by timepoint where significant at $p<0.05$. Additional mixed-effects models examined hourly patterns in sitting time, prolonged sitting time, and stepping time across the day according to work context.

RESULTS AND DISCUSSION

During waking hours, work-from-home days were associated with greater sitting time (+37.4 [95% CI:

24.0, 50.8] min/day), prolonged sitting time (+55.4 [35.9, 74.8] min/day), and longer mean sitting bout duration (+1.7 [0.8, 2.5] min/bout) relative to workplace days. These associations remained evident during working hours (+30.6 [20.5, 40.6] min/worktime sitting, +47.7 [33.6, 61.8] min/worktime prolonged sitting, and +7.8 [2.9, 12.6] min/bout). Stepping time during waking hours was lower on work-from-home than workplace days (-15.1 [-19.4, -10.7] min/day) and during working hours (-10.0 [-13.1, -6.9] min/worktime), with differences being attenuated over time. Hourly analyses suggested that the higher prolonged sitting observed on work-from-home days was driven primarily by differences observed during working hours.

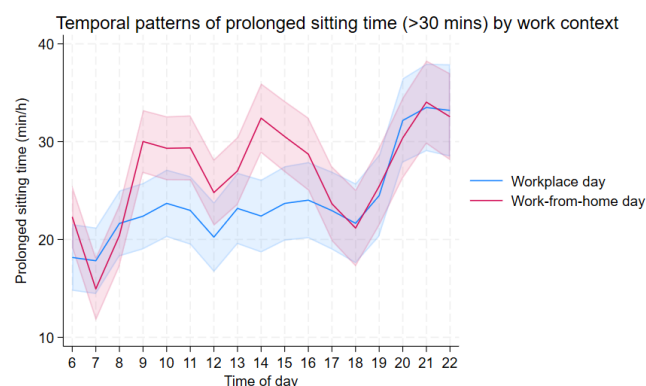


Figure 1 Hourly prolonged sitting time (>30 min bouts) during workplace and work-from-home days.

CONCLUSIONS

In office workers with type 2 diabetes, we showed that work-from-home may contribute to greater sitting, prolonged sitting, and lower stepping time, especially during working hours. As even modest differences in movement behaviours may be relevant to health outcomes, work-from-home settings may exacerbate the sedentary nature of desk-based work, representing a target for future interventions.